2 Forces and movement

2.1 Forces around us

We are going to:

- learn that forces can make things move and make things stop
- make predictions and say if they are right
- work in a safe way.

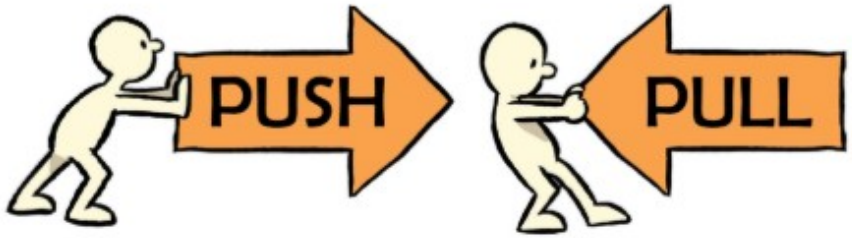
away	rocket
force	squashing
launch	stretch
pull/pulls	towards
push/pushes	



Getting started

- Tell a friend about things that you move at home.
- Do you have to **push** them or **pull** them to make them move?
- Show a friend something that moves when you push it.
- Now show them something that moves when you pull it.

Pushes and pulls are **forces**.



A push is a force **away** from you.

A pull is a force **towards** you.



This girl is pushing.



These boys are pulling.

Forces can make things move and they can make things stop.

This man uses a push to stop the ball.



Activity

Using forces to move things and stop things



You will need:
a ball, a bowl of water, a balloon, a chair and a pencil



How can you make a ball move?

Will the ball move if you push it?

Will it move if you pull it?

How can you make it stop?

Try making the other things move and making them stop.

Be careful of other people. Make sure you work in a safe way with the ball.

There are small forces and big forces.



Omar pulls his toys with elastic. The elastic can **stretch**.

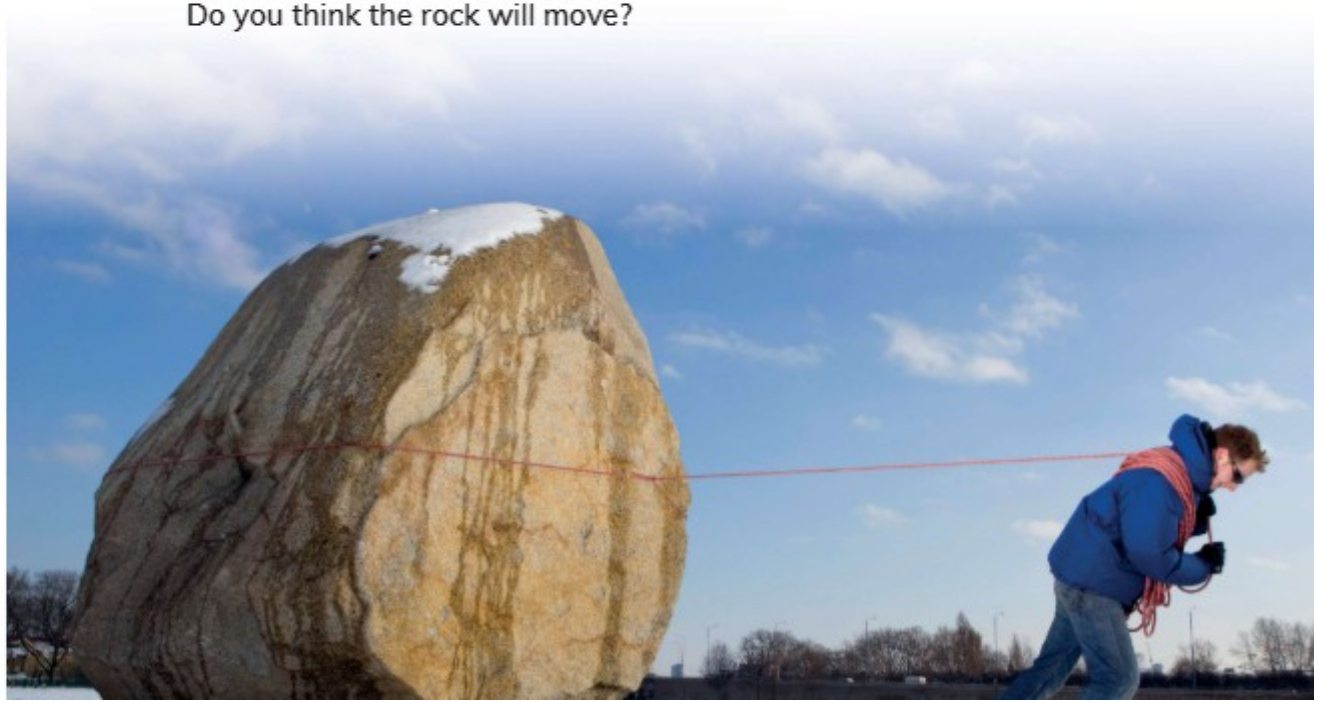
Questions

1 Which toy needs a big pull to make it move?

2 Which toy needs a small pull to make it move?

3 This man is pulling a big rock.

Do you think the rock will move?



Think like a scientist

Make a paper rocket launcher

You will need:
some paper and a plastic bottle



Zara has made a **rocket**. She can **launch** it by **squashing** the bottle with a push.

Make your own paper rocket.
Look at the pictures to see what to do.



1 2 3 4 5

Use a small push to launch your rocket. Then use a big push.

Launch your rocket from the floor.

Predict how high the rocket will go each time.
Use your hand to show how high you think your rocket will go.

What happens when you use a small push?

How is it different when you use a big push?

Were your predictions correct?

Remember to work safely. Do not launch your rocket towards people.

How am I doing?

Ask a friend to predict how high their rocket will go. Watch them launch their rocket. Can they say if their prediction was correct?

Which of your five senses can you use to observe forces?
Can you use more than one sense?

Look what I can do!

- ☐ I can say how a push or pull can make things move or make things stop.
- ☐ I can make predictions and say if they are correct.
- ☐ I can work in a safe way.

